MONAI-CARDIAC-FAT-FL • CLINICAL AI & FEDERATED LEARNING SaMD ARCHITECTURE

End-to-End Processing, Privacy Governance & Regulatory Pipeline

Multi-Vendor CT Ingestion → Dual-Mode Privacy Gate → 3D AI Engine → Local EHR/PACS & Multi-Center FL Consortium

FDA Class II 510(k) | IEC 62304 Class B | ISO 14971 FMEA | NVIDIA FLARE 2.5+ | OHIF v3.13 Next | HIPAA Safe Harbor

1. CT INGESTION

Universal 3D MetaTensor Loader

Multi-Vendor Scanners

Calibrated Volumetric Series

- Siemens Healthineers (Force, Flash)
- GE Healthcare (Revolution, HD)
- Philips Healthcare (Brilliance, iCT)
- Canon Medical (Aquilion ONE)
- Non-gated & Gated CCTA / CAC

Native 3D I/O Formats

Header & Geometry Preserving

- DICOM Series (Multi-file axial)
- NRRD (scan.nrrd / img.nrrd)
- NIfTI (.nii, .nii.gz) & MHA
- Natural Slice Sorting (IPP Z)
- Multi-series scout collision safe
- Fast os.scandir cohort discovery

Coordinate Preservation

Physical Patient Space

- **PixelSpacing (dx, dy mm)**
- **SliceThickness / Spacing (dz)**
- **ImagePositionPatient (x0, y0, z0)**
- **ImageOrientationPatient (cosines)**
- **Affine matrix alignment [4x4]**

Clinical

Research

2. DUAL-MODE PRIVACY GATE

Regulatory Trust Boundary Enforcement

CLINICAL MODE (--mode clinical)

Local Hospital Intranet Trust Zone

- Identified Patient Metadata Retained
- PatientName, MRN (PatientID)
- AccessionNumber, StudyDate, Physician
- Files directly to Hospital PACS archive
- Ingests into Patient EHR Chart (Epic)
- **RESTRICTION: Blocked from crossing** hospital enterprise firewall or FL

RESEARCH MODE (--mode research)

HIPAA Safe Harbor & RSNA V19 Profile

- Official RSNA Anonymizer (V19) Integration
  uv tool run --from rsna-anonymizer
- Native CLI: cardiac-fat-fl anonymize
- 18 HIPAA Safe Harbor Statutory PHI Scrubbed
- Pseudonyms: Participant {pid}, ACC_{pid}
- Cryptographic SHA-256 HMAC UID remapping
- Deterministic 30-day date shifting
- Pre-Flight PHI Audit Gate (deid-audit)
  Fail-closed security: Halts on raw PHI leak
- **Invariant: 100% Geometry & HU preserved**

3. 3D ANATOMY & FAT QUANTIFICATION

Unified Multi-Foundation & Geometric Engine

Selectable Chamber Backend (--chamber-backend)

Multi-Foundation Model Options

- **monai (Default)**
  NGC SegResNet 1.5mm isotropic, 104 labels
- **totalsegmentator**
  nnU-Net Task253, Bit-exact Harvard match
- **nvsegment / nvsegment_ct**
  NVIDIA-MedTech VISTA-3D foundation model
- **DeepHeartSeg ONNX**
  Zero-drift pericardial sac envelope (heart.seg.nrrd)

Clinical Invariants & 3D Spatial Transforms

Standardized Anatomical Taxonomy

- Adipose Attenuation Window: strictly [-190.0, -30.0] HU
- Coronary Sulci Derivation: geometric grooves (LAD, LCX, RCA)
- 10.0 mm Anisotropic Radial PVAT Dilation (Voronoi 3D EDT)
- CRISP-CT Vascular Inflammation Cutoff: Mean PVAT > -70.1 HU
- Empty Region Guardrail: 0.0 cm3 volume & NaN mean HU

8-Compartment Segmentation (eat_pvat.seg.nrrd)

Canonical 4-Way Synchronized Color Palette

EAT LA | EAT LV | EAT RA | EAT RV | PVAT LAD | PVAT LCX | PVAT RCA | EAT Total

Clinical Output

De-ID Model Gradients

PCCP Distilled Global Weights

LOCAL EHR, PACS & WORKSTATION

Hospital Clinical Production Deployment

Enterprise PACS Archive

DICOMweb & DIMSE Services

- DICOM-SEG (SOP .66.4, 8 layers)
- DICOM-SR TID 1500 (SOP .88.34)
- Encapsulated PDF (SOP .104.1)
- DICOMweb STOW-RS (TLS 1.3)
- DIMSE C-STORE (pynetdicom)
- Custom Enterprise Root CAs

Hospital EHR & OHIF

HL7 FHIR & Zero-Footprint UI

- HL7 FHIR R4 DiagnosticReport
- DocumentReference Chart PDF
- LOINC 84245-0 EAT volume cm3
- RadLex RID49533 PVAT HU
- Open.epic MCP Server (stdio)
- OHIF v3.13 2x2 Slicer Workstation

(WORLD) FEDERATED LEARNING

Privacy-Preserving Multi-Hospital AI Consortium

NVIDIA FLARE 2.5+

Zero DICOM Transfer Across Firewalls

- **Zero CT Image / Patient Transfer**
- FedProx Proximal Regularization
- FedBN: Local Normalization (BatchNorm3d remains on-site)
- CrossSiteModelEval Consensus
- Global weights: SRV_FL_model.pt
- Self-healing Windows patching

Federated Analytics

Welford Scalar Moment Aggregation

- Exact Welford Summary Moments (N, sum x, sum x^2, min, max)
- 100% Mathematical Equivalence to pooled multi-center data
- Zero patient re-identification
- Real-Time FL Monitor Dashboard
- Automated site loss/DICE curves

FOUNDATION FOR REGULATORY FILING: FDA Class II 510(k) Software as a Medical Device (SaMD)

Comprehensive Quality Management System (QMS), IEC 62304 Life Cycle, ISO 14971 Risk Management & AI/ML PCCP

Pillar 1: FDA 510(k) Device Profile

- Pathway: 21 CFR § 892.2050
- Product Codes: LLZ (Image Proc), QIH (Automated Quant)
- Documentation Level: Basic (Radiologist-in-the-loop)
- Reference Predicate: Harvard CardiacFatQuant
  Controlled Parity: r = 1.000, 100% DICE bit-exact
- Indications: 3D EAT/PVAT volumetry from cardiac CT

Pillar 2: IEC 62304 Software Safety

- Safety Class: Class B (Non-serious injury risk)
- Traceability Matrix: REQ-001 through REQ-013
- Verification Suite: 146 automated pytest tests
- Turnkey MAP Application Package (monai_deploy.py)
- Fail-Closed Execution Architecture
- Header-Preserving Spatial & Coordinate Integrity

Pillar 3: ISO 14971 Risk Controls

- FMEA Hazard Analysis: HAZ-001 through HAZ-010
- Blooming Mitigation: Strict -30 HU contrast threshold
- Boundary Guardrail: OHIF mandatory contour review
- Anisotropic Sampling: Physical mm distance EDT
- Model Tampering Defense: SHA-256 cryptographic check
- PHI Leak Defense: RSNA V19 & pre-flight deid-audit

Pillar 4: FDA AI/ML PCCP Protocol

- Predetermined Change Control Plan (FDA 2024)
- Post-Market Federated Learning Retraining Bounds
- Invariants Frozen: 8-ch, [-190, -30] HU, 10mm EDT
- Multi-Site Acceptance: Dice >= 0.90, Pearson r >= 0.98
- Automated Rollback: Reverts on threshold defect
- Section 524B Cybersecurity: CycloneDX SBOM & SAST